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1) Ferritic/Martensitic Steels or other materials are being considered and evaluated for in-reactor performance due to excessive swelling exhibited by currently-implemented austenitic stainless steels. Optimization of the F/M composition is crucial as material and mechanical properties are highly influenced by the quantities and ratio of ferrite and martensite. Generally, the ferrite has a strengthening characteristic to the martensite, and increase the ductility. However, excessive amounts can increase susceptibility to void formation, or lead to excessive precipitate formation leading to less favorable properties. -needed more. Cr effects?

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2) Void formation and associated swelling in F/M steels is significantly lower than in austenitic steels. Recombination and the presence of interstitials or precipitates significantly reduces the overall swelling behavior of F/M steels depending on the composition. This swelling 'resistance' with irradiation is also dependent on dose and fluence, as high fluence and higher dose materials may still exhibit some swelling, albeit reduced. -needed more...

7/10

3) The ductile-brittle transition temperature is a fracture property describing the transition from the brittle lower shelf-regime of the fracture toughness, to the more ductile upper shelf regime. This temperature typically increases with irradiation making some materials more susceptible to fracture at reactor conditions at higher dose/dpa. Embrittlement in F/M primarily comes from the formation of voids, thus by increasing the transition temperature. Void sinks in F/M steels can be prevalent with higher quantities of martensite at higher irradiation dose. 7/10

4) FeCrAl alloys are of particular interest in advanced cladding because of the incredible swelling and corrosion resistance at higher dpa and irradiation temperature. Balancing the quantities of Fe &Cr? and Al play key roles in limiting precipitate formation. Formation of intermetallic phases with Al can also occur, thus Zr may sometimes be added as a stabilizing element. Cr content (up to 20-25%) aids with the corrosion resistance significantly as well.

-almost there

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5) The YO_2 particles in ODS steels are the primary mechanism behind the oxide dispersion and strengthening aspects in ODS steels. The particles act as a safeguard against further swelling, and also can increase mechanical performance of the system as particles can absorb some of the irradiation damage, via defects and voids, which in turn can "self-heal" ~~and~~ possibly due to recombination effects. This makes them resistant to significant change from irradiation

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6) Corrosion resistance of nickel-based alloys is a massive benefit in SFRs or other advanced reactors, However, He and H_2 generation stemming from ^{58}Ni can cause significant embrittlement concerns and challenges. He and H_2 generation occurs via transmuting effects during neutron irradiation and is a limiting phenomenon when it comes to usefulness. This drives excessive intergranular bubble formations as well as void formations that can also be sources of hydrogen gas accumulation.

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- high T mech. properties

7) U-Si fuels have a much higher Uranium density than UAlx plate fuels do. However, maintaining exact stoichemetric U-Si fuels is near-impossible leading to an excess of Si. Due to this fission product generation and gas mobility is excessive in U-Si fuels, as the structure can be amorphous. U3Si and U3Si2 exhibit slightly different swelling characteristics, with U3Si2 having additional interactions ~~with~~ the aluminum matrix, restricting the swelling slightly when compared to U_3Si.

10/12

- amorphous phase differences
- good UAlx?

8) The gamma-phase of U-Mo plate fuel is necessary and desired for fuel stability, thus in fuel fabrication, the plate fuel is heat treated to contain the gamma phase. In-reactor, the gamma content is somewhat retained due to the swelling behavior of the fuel in the slow-generation regime.

- swelling is unrelated

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9) The Solidus/liquidus gap can significantly influence the homogeneous distribution of Mo throughout the system. Formation of Mo-rich as well as depleted zones are possible and form within the fuel. It also can contribute to the formation of interaction layers between the aluminum and fuel particulate within the plate.

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10) Two primary regimes for fission product generation exist in U-Mo plate fuel: a low-dose, low generation regime, and a high-dose, high generation regime. The transition region between these depends heavily on the quantity of gamma phase of U-Mo that is present. Generally, the low-dose regime dominates initially until the voids generated begin to "self-heal" slightly and the gamma content is returned to near as-fabricated condition, after which rapid fission product generation dominates. - missed the main concept

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11) The interaction layer (IL) how? in U-Mo fuels plays a huge role in fuel lifetime, fuel dispersion and uranium density, and fuel swelling. Monolithic U-Mo fuel forms have been developed as a primary mitigation strategy. Addition of Zr has also been used to prevent formation of Al-based phases in the fuel. One plate fuel form segregates the fuel region from the Al matrix using a Zr foil, though fabrication of these plates remains a challenge. 6/7

6/6

12) Al is ideally suited for a research reactor because it is cheap, easily machinable, corrosion resistant at relevant temperatures for research reactors (~ 100-200 °C). It also has a lower neutron cross section, enabling a higher neutron flux in desirable locations. Al is less usable in LWRs because of its strength and corrosion resistance at LWR temperatures (T ~ 300-400 °C), which are higher than research reactors.

- melting point